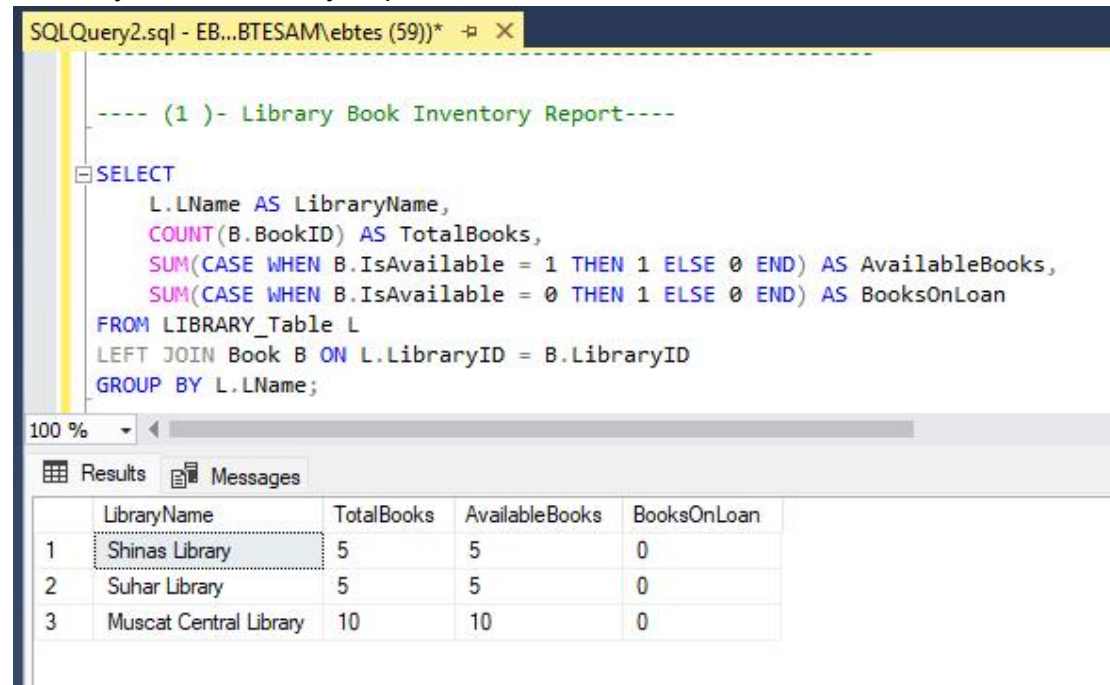


# Section 1: Complex Queries with Joins

## 1. Library Book Inventory Report:



The screenshot shows a SQL query window titled "SQLQuery2.sql - EB...BTESAM\ebtes (59))". The query is a SELECT statement that generates a library book inventory report. It uses a LEFT JOIN to combine the LIBRARY\_Table (L) with the Book table (B) based on LibraryID. The query calculates the total number of books, the number of available books, and the number of books on loan for each library.

```
----- (1 )- Library Book Inventory Report-----  
  
SELECT  
    L.LName AS LibraryName,  
    COUNT(B.BookID) AS TotalBooks,  
    SUM(CASE WHEN B.IsAvailable = 1 THEN 1 ELSE 0 END) AS AvailableBooks,  
    SUM(CASE WHEN B.IsAvailable = 0 THEN 1 ELSE 0 END) AS BooksOnLoan  
FROM LIBRARY_Table L  
LEFT JOIN Book B ON L.LibraryID = B.LibraryID  
GROUP BY L.LName;
```

The Results tab shows the following data:

|   | LibraryName            | TotalBooks | AvailableBooks | BooksOnLoan |
|---|------------------------|------------|----------------|-------------|
| 1 | Shinas Library         | 5          | 5              | 0           |
| 2 | Suhar Library          | 5          | 5              | 0           |
| 3 | Muscat Central Library | 10         | 10             | 0           |

## 2. Active Borrowers Analysis

SQLQuery2.sql - EB...BTESAM\ebtes (59)\*

```
----- (2) Active Borrowers Analysis-----  
  
SELECT  
    M.FullName AS MemberName,  
    M.Email,  
    B.Title AS BookTitle,  
    L.LoanDate,  
    L.DueDate,  
    L.Status  
FROM Loan_table L  
JOIN MEMBER_TABLE M ON L.MemberID = M.MemberID  
JOIN Book B ON L.BookID = B.BookID  
WHERE L.Status IN ('Issued', 'Overdue');
```

100 %

Results Messages

|   | MemberName | Email              | BookTitle           | LoanDate   | DueDate    | Status |
|---|------------|--------------------|---------------------|------------|------------|--------|
| 1 | EBTESAM    | EBTESAM@GOOGLE.com | Data Science Basics | 2025-12-01 | 2025-12-15 | Issued |
| 2 | AHMED      | AHMED@OUTLOOK.com  | Fairy Tales         | 2025-12-05 | 2025-12-20 | Issued |
| 3 | ALI        | ALI@gmail.com      | Data Science Basics | 2025-04-01 | 2025-04-15 | Issued |
| 4 | SARA       | SARA@gmail.com     | SQL Mastery         | 2025-05-03 | 2025-04-18 | Issued |
| 5 | MOHAMMED   | MOHAMMED@gmail.com | Fairy Tales         | 2025-02-01 | 2025-08-13 | Issued |
| 6 | HIND       | HIND@gmail.com     | SQL Mastery         | 2025-07-08 | 2025-07-12 | Issued |
| 7 | KHALID     | KHALID@gmail.com   | Data Science Basics | 2025-03-09 | 2025-09-11 | Issued |
| 8 | AISHA      | AISHA@gmail.com    | Data Science Basics | 2025-06-03 | 2025-10-10 | Issued |

### 3. Overdue Loans with Member Details:

SQLQuery2.sql - EB...BTESAM\ebtes (59))\*

----- (3) Issued Loans with Member Details -----

SELECT

```
M.FullName AS MemberName,  
M.PhoneNumber,  
B.Title AS BookTitle,  
LT.LName AS LibraryName,  
DATEDIFF(DAY, L.DueDate, GETDATE()) AS DaysOverdue,  
ISNULL(SUM(P.Amount), 0) AS TotalFinePaid  
FROM Loan_table L  
JOIN MEMBER_TABLE M ON L.MemberID = M.MemberID  
JOIN Book B ON L.BookID = B.BookID  
JOIN LIBRARY_Table LT ON B.LibraryID = LT.LibraryID  
LEFT JOIN Payment P ON L.LoanID = P.LoanID  
WHERE L.Status = 'Issued'  
GROUP BY  
M.FullName, M.PhoneNumber, B.Title, LT.LName, L.DueDate;
```

100 %

Results Messages

|   | MemberName | PhoneNumber | BookTitle           | LibraryName    | DaysOverdue | TotalFinePaid |
|---|------------|-------------|---------------------|----------------|-------------|---------------|
| 1 | AHMED      | 111-5678    | Fairy Tales         | Shinas Library | 10          | 6.00          |
| 2 | AISHA      | 900-6666    | Data Science Basics | Shinas Library | 81          | 0.00          |
| 3 | ALI        | 900-1111    | Data Science Basics | Shinas Library | 259         | 0.00          |
| 4 | EBTESAM    | 333-1234    | Data Science Basics | Shinas Library | 15          | 0.00          |
| 5 | HIND       | 900-4444    | SQL Mastery         | Shinas Library | 171         | 0.00          |
| 6 | KHALID     | 900-5555    | Data Science Basics | Shinas Library | 110         | 8.50          |
| 7 | MOHAMMED   | 900-3333    | Fairy Tales         | Shinas Library | 139         | 0.00          |
| 8 | SARA       | 900-2222    | SQL Mastery         | Shinas Library | 256         | 0.00          |

#### 4. Staff Performance Overview:

SQLQuery2.sql - EB...BTESAM\ebtes (59))\*

----(4) Staff Performance Overview-----

```
SELECT
  LT.LName AS LibraryName,
  S.FullName AS StaffName,
  S.Position,
  COUNT(B.BookID) AS BooksManaged
FROM LIBRARY_Table LT
JOIN STAFF_TABLE S ON LT.LibraryID = S.LibraryID
LEFT JOIN Book B ON LT.LibraryID = B.LibraryID
GROUP BY
  LT.LName, S.FullName, S.Position;
```

100 %

Results Messages

|   | LibraryName    | StaffName       | Position            | BooksManaged |
|---|----------------|-----------------|---------------------|--------------|
| 1 | Shinas Library | EBTESAM ALZAABI | Librarian           | 5            |
| 2 | Shinas Library | NURA ALHOSANI   | Assistant Librarian | 5            |
| 3 | Suhar Library  | AMANI ALSALHI   | Auditor             | 5            |

### 5. Book Popularity Report:

SQLQuery2.sql - EB...BTESAM\ebtes (59))\*

```
---(5) Book Popularity Report ---  
  
SELECT  
  B.Title,  
  B.ISBN,  
  B.Genre,  
  COUNT(L.LoanID) AS TimesLoaned,  
  AVG(R.Rating) AS AverageRating  
FROM Book B  
JOIN Loan_table L ON B.BookID = L.BookID  
LEFT JOIN Review R ON B.BookID = R.BookID  
GROUP BY  
  B.Title, B.ISBN, B.Genre  
HAVING COUNT(L.LoanID) >= 3;
```

100 %

Results Messages

|   | Title               | ISBN     | Genre       | TimesLoaned | AverageRating |
|---|---------------------|----------|-------------|-------------|---------------|
| 1 | Data Science Basics | 111-0001 | Non-fiction | 4           | NULL          |
| 2 | SQL Mastery         | 111-0002 | Reference   | 3           | NULL          |
| 3 | Fairy Tales         | 111-0003 | Children    | 3           | NULL          |

-- This query generates a report of the most popular books in the library database. Popularity is measured by the number of times a book has been borrowed. Additionally, it calculates the average rating of each book based on member reviews.

Average rating uses **COALESCE** to handle books without reviews.

## 6. Member Reading History:

SQLQuery2.sql - EB...BTESAM\ebtes (59))\*

---(6) Member Reading History-----

SELECT

M.FullName AS MemberName,  
B.Title AS BookTitle,  
L.LoanDate,  
L.ReturnDate,  
R.Rating,  
R.Comments

FROM MEMBER\_TABLE M

LEFT JOIN Loan\_table L ON M.MemberID = L.MemberID

LEFT JOIN Book B ON L.BookID = B.BookID

LEFT JOIN Review R  
ON R.MemberID = M.MemberID  
AND R.BookID = B.BookID

ORDER BY  
M.FullName, L.LoanDate;

100 %

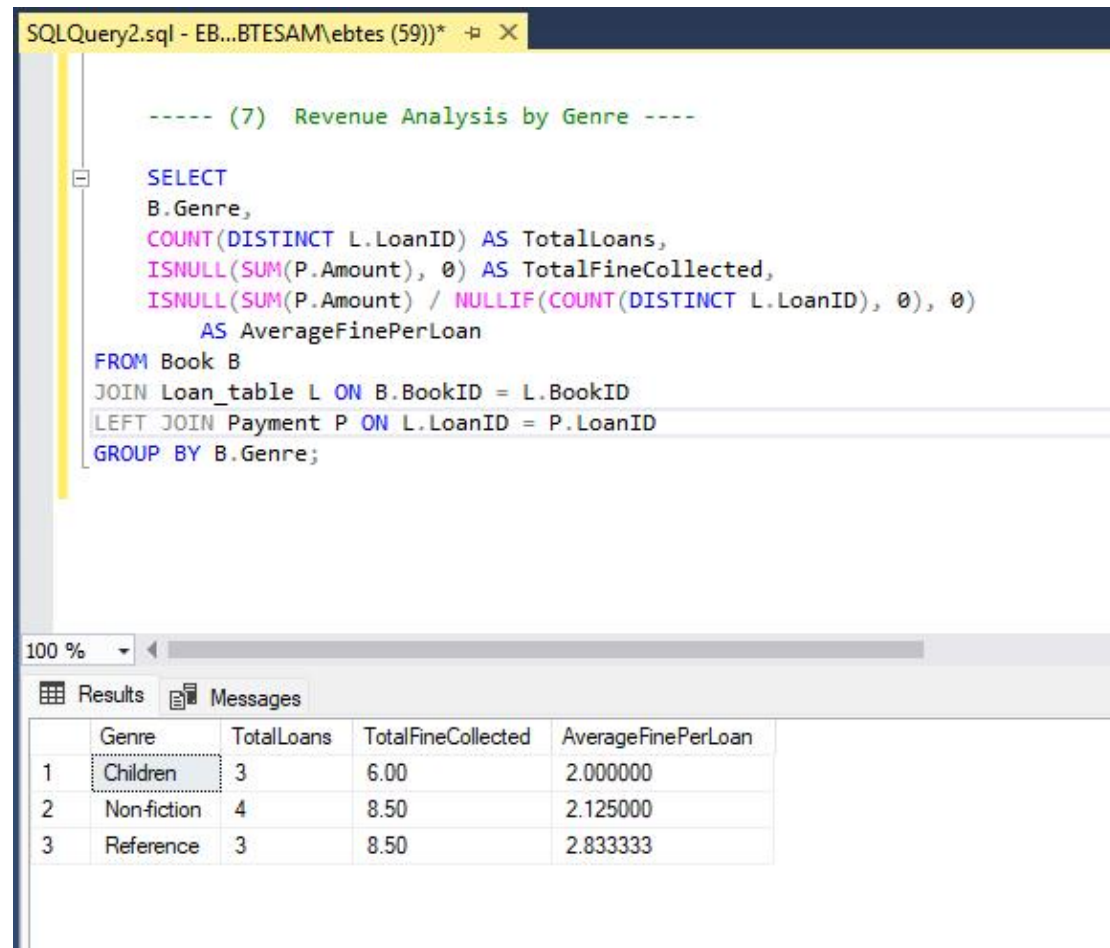
Results Messages

|    | MemberName | Book Title          | LoanDate   | ReturnDate | Rating | Comments |
|----|------------|---------------------|------------|------------|--------|----------|
| 1  | AHMED      | Fairy Tales         | 2025-12-05 | NULL       | NULL   | NULL     |
| 2  | AISHA      | Data Science Basics | 2025-06-03 | NULL       | NULL   | NULL     |
| 3  | ALI        | Data Science Basics | 2025-04-01 | NULL       | NULL   | NULL     |
| 4  | EBTESAM    | Data Science Basics | 2025-12-01 | NULL       | NULL   | NULL     |
| 5  | EBTESAM    | SQL Mastery         | 2025-12-02 | 2025-12-14 | NULL   | NULL     |
| 6  | FATMA      | NULL                | NULL       | NULL       | NULL   | NULL     |
| 7  | HIND       | SQL Mastery         | 2025-07-08 | NULL       | NULL   | NULL     |
| 8  | KHALID     | Data Science Basics | 2025-03-09 | NULL       | NULL   | NULL     |
| 9  | MOHAMMED   | Fairy Tales         | 2025-02-01 | NULL       | NULL   | NULL     |
| 10 | SARA       | SQL Mastery         | 2025-05-03 | NULL       | NULL   | NULL     |
| 11 | YOUSEF     | Fairy Tales         | 2025-01-05 | 2025-04-18 | NULL   | NULL     |

This query shows the complete borrowing history of each library member, including both currently borrowed and previously returned books. It also includes any reviews they left for the books.



## 7. Revenue Analysis by Genre:



The screenshot displays a SQL query window titled "SQLQuery2.sql - EB...BTESAM\ebtes (59))". The query is a SELECT statement that calculates revenue metrics by book genre. It joins the Book table (B) with the Loan table (L) and the Payment table (P). The results are grouped by genre.

```
----- (7) Revenue Analysis by Genre -----  
  
SELECT  
    B.Genre,  
    COUNT(DISTINCT L.LoanID) AS TotalLoans,  
    ISNULL(SUM(P.Amount), 0) AS TotalFineCollected,  
    ISNULL(SUM(P.Amount) / NULLIF(COUNT(DISTINCT L.LoanID), 0), 0)  
        AS AverageFinePerLoan  
FROM Book B  
JOIN Loan_table L ON B.BookID = L.BookID  
LEFT JOIN Payment P ON L.LoanID = P.LoanID  
GROUP BY B.Genre;
```

The results are shown in a table with the following data:

|   | Genre       | TotalLoans | TotalFineCollected | AverageFinePerLoan |
|---|-------------|------------|--------------------|--------------------|
| 1 | Children    | 3          | 6.00               | 2.000000           |
| 2 | Non-fiction | 4          | 8.50               | 2.125000           |
| 3 | Reference   | 3          | 8.50               | 2.833333           |

This query calculates the total fines collected per book genre, total number of loans for each genre, and the average fine per loan. It helps the library understand which genres generate the most revenue from fines.

## Section 2: Aggregate Functions and Grouping

### 8. Monthly Loan Statistics:

SQLQuery2.sql - EB...BTESAM\ebtes (59))

```
----- Section 2: Aggregate Functions and Grouping-----  
  
----- 8.Monthly Loan Statistics-----  
  
SELECT  
    DATENAME(MONTH, LoanDate) AS MonthName,  
    COUNT(*) AS TotalLoans,  
    SUM(CASE WHEN Status = 'Returned' THEN 1 ELSE 0 END) AS TotalReturned,  
    SUM(CASE WHEN Status IN ('Issued', 'Overdue') THEN 1 ELSE 0 END) AS StillIssuedOrOverdue  
FROM Loan_table  
WHERE YEAR(LoanDate) = YEAR(GETDATE())  
GROUP BY  
    DATENAME(MONTH, LoanDate),  
    MONTH(LoanDate)  
ORDER BY MONTH(LoanDate);
```

100 %

Results Messages

|   | MonthName | TotalLoans | TotalReturned | StillIssuedOrOverdue |
|---|-----------|------------|---------------|----------------------|
| 1 | January   | 1          | 1             | 0                    |
| 2 | February  | 1          | 0             | 1                    |
| 3 | March     | 1          | 0             | 1                    |
| 4 | April     | 1          | 0             | 1                    |
| 5 | May       | 1          | 0             | 1                    |
| 6 | June      | 1          | 0             | 1                    |
| 7 | July      | 1          | 0             | 1                    |
| 8 | December  | 3          | 1             | 2                    |



## 9. Member Engagement Metrics:

SQLQuery2.sql - EB...BTESAM\ebtes (59))

```
-----9. Member Engagement Metrics:-----  
  
SELECT  
    M.FullName AS MemberName,  
    COUNT(L.LoanID) AS TotalBooksBorrowed,  
    SUM(CASE WHEN L.Status IN ('Issued', 'Overdue') THEN 1 ELSE 0 END)  
        AS CurrentlyOnLoan,  
    ISNULL(SUM(P.Amount), 0) AS TotalFinesPaid,  
    AVG(R.Rating) AS AverageRatingGiven  
FROM MEMBER_TABLE M  
JOIN Loan_table L ON M.MemberID = L.MemberID  
LEFT JOIN Payment P ON L.LoanID = P.LoanID  
LEFT JOIN Review R ON M.MemberID = R.MemberID  
GROUP BY  
    M.FullName;
```

100 %

Results Messages

|   | MemberName | TotalBooksBorrowed | CurrentlyOnLoan | TotalFinesPaid | AverageRatingGiven |
|---|------------|--------------------|-----------------|----------------|--------------------|
| 1 | AHMED      | 1                  | 1               | 6.00           | NULL               |
| 2 | AISHA      | 1                  | 1               | 0.00           | NULL               |
| 3 | ALI        | 1                  | 1               | 0.00           | NULL               |
| 4 | EBTESAM    | 3                  | 1               | 8.50           | NULL               |
| 5 | HIND       | 1                  | 1               | 0.00           | NULL               |
| 6 | KHALID     | 1                  | 1               | 8.50           | NULL               |
| 7 | MOHAMMED   | 1                  | 1               | 0.00           | NULL               |
| 8 | SARA       | 1                  | 1               | 0.00           | NULL               |
| 9 | YOUSEF     | 1                  | 0               | 0.00           | NULL               |

## 10. Library Performance Comparison

SQLQuery2.sql - EB...BTESAM\ebtes (59)\*

-----10.Library Performance Comparison-----

```
SELECT
    LT.LName AS LibraryName,
    COUNT(DISTINCT B.BookID) AS TotalBooksOwned,
    COUNT(DISTINCT L.MemberID) AS ActiveMembers,
    ISNULL(SUM(P.Amount), 0) AS TotalFineRevenue,
    CAST(COUNT(DISTINCT B.BookID) * 1.0
        / NULLIF(COUNT(DISTINCT L.MemberID), 0)
        AS DECIMAL(10,2)) AS AvgBooksPerMember
FROM LIBRARY_Table LT
LEFT JOIN Book B ON LT.LibraryID = B.LibraryID
LEFT JOIN Loan_table L ON B.BookID = L.BookID
LEFT JOIN Payment P ON L.LoanID = P.LoanID
GROUP BY
    LT.LName;
```

100 %

Results Messages

|   | LibraryName            | TotalBooksOwned | ActiveMembers | TotalFineRevenue | AvgBooksPerMember |
|---|------------------------|-----------------|---------------|------------------|-------------------|
| 1 | Muscat Central Library | 10              | 0             | 0.00             | NULL              |
| 2 | Shinas Library         | 5               | 9             | 23.00            | 0.56              |
| 3 | Suhar Library          | 5               | 0             | 0.00             | NULL              |

## 11. High-Value Books Analysis

SQLQuery2.sql - EB...BTESAM\ebtes (59))\*

```
-----11.High-Value Books Analysis-----  
  
SELECT  
    B.Title,  
    B.Genre,  
    B.Price,  
    AVG(B.Price) OVER (PARTITION BY B.Genre) AS GenreAveragePrice,  
    B.Price - AVG(B.Price) OVER (PARTITION BY B.Genre) AS PriceDifference  
FROM Book B  
WHERE B.Price >  
    (SELECT AVG(B2.Price)  
     FROM Book B2  
     WHERE B2.Genre = B.Genre);
```

100 %

Results Messages

|    | Title              | Genre       | Price | GenreAveragePrice | PriceDifference |
|----|--------------------|-------------|-------|-------------------|-----------------|
| 1  | Fairy Tales        | Children    | 12.00 | 11.750000         | 0.250000        |
| 2  | Children Fun Book  | Children    | 11.50 | 11.750000         | -0.250000       |
| 3  | Modern Fiction     | Fiction     | 20.00 | 19.000000         | 1.000000        |
| 4  | Mystery Island     | Fiction     | 18.00 | 19.000000         | -1.000000       |
| 5  | AI Future          | Non-fiction | 28.00 | 28.000000         | 0.000000        |
| 6  | History of Oman    | Non-fiction | 26.00 | 28.000000         | -2.000000       |
| 7  | Tech Trends        | Non-fiction | 30.00 | 28.000000         | 2.000000        |
| 8  | Advanced SQL       | Reference   | 48.00 | 47.666666         | 0.333334        |
| 9  | Encyclopedia Vol 1 | Reference   | 50.00 | 47.666666         | 2.333334        |
| 10 | Science Explained  | Reference   | 45.00 | 47.666666         | -2.666666       |

## 12. Payment Pattern Analysis:

SQLQuery2.sql - EB...BTESAM\ebtes (59)) \* -> X

-----12. Payment Pattern Analysis-----

```
SELECT
    Method AS PaymentMethod,
    COUNT(*) AS NumberOfTransactions,
    SUM(Amount) AS TotalCollected,
    AVG(Amount) AS AveragePaymentAmount,
    CAST(
        SUM(Amount) * 100.0 / SUM(SUM(Amount)) OVER ()
        AS DECIMAL(5,2)
    ) AS PercentageOfTotalRevenue
FROM Payment
GROUP BY Method;
```

100 %

Results Messages

|   | PaymentMethod | NumberOfTransactions | TotalCollected | AveragePaymentAmount | PercentageOfTotalRevenue |
|---|---------------|----------------------|----------------|----------------------|--------------------------|
| 1 | Card          | 2                    | 12.00          | 6.000000             | 52.17                    |
| 2 | Cash          | 2                    | 11.00          | 5.500000             | 47.83                    |

# Section 3: Views Creation

## 13. vw\_CurrentLoans

SQLQuery2.sql - EB...BTESAM\ebtes (59) \* X

```
-----13.vw_CurrentLoans-----
CREATE VIEW vw_CurrentLoans AS
SELECT
    L.LoanID,
    M.MemberID,
    M.FullName AS MemberName,
    M.Email,
    M.PhoneNumber,
    B.BookID,
    B.Title AS BookTitle,
    B.Genre,
    B.ShelfLocation,
    L.LoanDate,
    L.DueDate,
    L.Status,
    DATEDIFF(DAY, GETDATE(), L.DueDate) AS DaysUntilDue,
    CASE
        WHEN L.DueDate < GETDATE() THEN DATEDIFF(DAY, L.DueDate, GETDATE())
        ELSE 0
    END AS DaysOverdue
FROM Loan_table L
JOIN MEMBER_TABLE M ON L.MemberID = M.MemberID
JOIN Book B ON L.BookID = B.BookID
WHERE L.Status IN ('Issued', 'Overdue');

SELECT * FROM vw_CurrentLoans;
```

100 %

|   | LoanID | MemberID | MemberName | Email              | PhoneNumber | BookID | BookTitle           | Genre       | ShelfLocation | LoanDate   | DueDate    | Status | DaysUntilDue | DaysOverdue |
|---|--------|----------|------------|--------------------|-------------|--------|---------------------|-------------|---------------|------------|------------|--------|--------------|-------------|
| 1 | 1      | 1        | EBTESAM    | EBTESAM@GOOGLE.com | 333-1234    | 1      | Data Science Basics | Non-fiction | A2            | 2025-12-01 | 2025-12-15 | Issued | -15          | 15          |
| 2 | 3      | 2        | AHMED      | AHMED@OUTLOOK.com  | 111-5678    | 3      | Fairy Tales         | Children    | A4            | 2025-12-05 | 2025-12-20 | Issued | -10          | 10          |
| 3 | 4      | 4        | ALI        | ALI@gmail.com      | 900-1111    | 1      | Data Science Basics | Non-fiction | A2            | 2025-04-01 | 2025-04-15 | Issued | -259         | 259         |
| 4 | 5      | 5        | SARA       | SARA@gmail.com     | 900-2222    | 2      | SQL Mastery         | Reference   | A3            | 2025-05-03 | 2025-04-18 | Issued | -256         | 256         |
| 5 | 6      | 6        | MOHAMMED   | MOHAMMED@gmail.com | 900-3333    | 3      | Fairy Tales         | Children    | A4            | 2025-02-01 | 2025-08-13 | Issued | -139         | 139         |
| 6 | 7      | 7        | HIND       | HIND@gmail.com     | 900-4444    | 2      | SQL Mastery         | Reference   | A3            | 2025-07-08 | 2025-07-12 | Issued | -171         | 171         |
| 7 | 8      | 8        | KHALID     | KHALID@gmail.com   | 900-5555    | 1      | Data Science Basics | Non-fiction | A2            | 2025-03-09 | 2025-09-11 | Issued | -110         | 110         |
| 8 | 9      | 9        | AISHA      | AISHA@gmail.com    | 900-6666    | 1      | Data Science Basics | Non-fiction | A2            | 2025-06-03 | 2025-10-10 | Issued | -81          | 81          |

## 14. Vw\_Library Statistics

SQLQuery2.sql - EB...BTESAM\ebtes (59) \* X

```
-----14. Vw_Library Statistics-----
CREATE VIEW vw_LibraryStatistics AS
SELECT
    Lib.LibraryID,
    Lib.LName AS LibraryName,
    COUNT(DISTINCT B.BookID) AS TotalBooks,
    SUM(CASE WHEN B.IsAvailable = 1 THEN 1 ELSE 0 END) AS AvailableBooks,
    COUNT(DISTINCT M.MemberID) AS TotalMembers,
    SUM(CASE WHEN L.Status IN ('Issued', 'Overdue') THEN 1 ELSE 0 END) AS ActiveLoans,
    COUNT(DISTINCT S.StaffID) AS TotalStaff,
    ISNULL(SUM(P.Amount), 0) AS TotalRevenue
FROM LIBRARY_Table Lib
LEFT JOIN Book B ON B.LibraryID = Lib.LibraryID
LEFT JOIN STAFF_TABLE S ON S.LibraryID = Lib.LibraryID
LEFT JOIN Loan_table L ON L.BookID = B.BookID
LEFT JOIN Payment P ON P.LoanID = L.LoanID
LEFT JOIN MEMBER_TABLE M ON M.MemberID = L.MemberID
GROUP BY Lib.LibraryID, Lib.LName;

SELECT * FROM vw_LibraryStatistics;
```

100 %

|   | LibraryID | LibraryName            | TotalBooks | AvailableBooks | TotalMembers | ActiveLoans | TotalStaff | TotalRevenue |
|---|-----------|------------------------|------------|----------------|--------------|-------------|------------|--------------|
| 1 | 1         | Shinas Library         | 5          | 26             | 9            | 16          | 2          | 46.00        |
| 2 | 2         | Suhar Library          | 5          | 5              | 0            | 0           | 1          | 0.00         |
| 3 | 3         | Muscat Central Library | 10         | 10             | 0            | 0           | 0          | 0.00         |

## 15. Vw\_Book Details With Reviews

SQL Query2.sql - EB...BTESAM\ebtes (59))

```

-----15.Vw_Book Details With Reviews-----
CREATE VIEW vw_BookDetailsWithReviews AS
SELECT
    B.BookID,
    B.ISBN,
    B.Title,
    B.Genre,
    B.Price,
    B.ShelfLocation,
    B.IsAvailable,
    Lib.LName AS LibraryName,
    COUNT(R.ReviewID) AS TotalReviews,
    ISNULL(AVG(CAST(R.Rating AS DECIMAL(3,2))),0) AS AverageRating,
    MAX(R.ReviewDate) AS LatestReviewDate
FROM Book B
LEFT JOIN Review R ON R.BookID = B.BookID
LEFT JOIN LIBRARY_Table Lib ON B.LibraryID = Lib.LibraryID
GROUP BY
    B.BookID, B.ISBN, B.Title, B.Genre, B.Price, B.ShelfLocation, B.IsAvailable, Lib.LName;

SELECT * FROM vw_BookDetailsWithReviews;

```

100 %

Results Messages

|    | BookID | ISBN     | Title               | Genre       | Price | ShelfLocation | IsAvailable | LibraryName            | TotalReviews | AverageRating | LatestReviewDate |
|----|--------|----------|---------------------|-------------|-------|---------------|-------------|------------------------|--------------|---------------|------------------|
| 1  | 11     | 333-0001 | Modern Fiction      | Fiction     | 20.00 | C1            | 1           | Muscat Central Library | 0            | 0.000000      | NULL             |
| 2  | 12     | 333-0002 | Learning Python     | Reference   | 42.00 | C2            | 1           | Muscat Central Library | 0            | 0.000000      | NULL             |
| 3  | 13     | 333-0003 | Healthy Living      | Non-fiction | 19.00 | C3            | 1           | Muscat Central Library | 0            | 0.000000      | NULL             |
| 4  | 14     | 333-0004 | Children Fun Book   | Children    | 11.50 | C4            | 1           | Muscat Central Library | 0            | 0.000000      | NULL             |
| 5  | 15     | 333-0005 | Advanced SQL        | Reference   | 48.00 | C5            | 1           | Muscat Central Library | 0            | 0.000000      | NULL             |
| 6  | 16     | 333-0006 | Short Stories       | Fiction     | 14.00 | C6            | 1           | Muscat Central Library | 0            | 0.000000      | NULL             |
| 7  | 17     | 333-0007 | History of Oman     | Non-fiction | 26.00 | C7            | 1           | Muscat Central Library | 0            | 0.000000      | NULL             |
| 8  | 18     | 333-0008 | Math for Kids       | Children    | 9.50  | C8            | 1           | Muscat Central Library | 0            | 0.000000      | NULL             |
| 9  | 19     | 333-0009 | Tech Trends         | Non-fiction | 30.00 | C9            | 1           | Muscat Central Library | 0            | 0.000000      | NULL             |
| 10 | 20     | 333-0010 | Fantasy World       | Fiction     | 17.00 | C10           | 1           | Muscat Central Library | 0            | 0.000000      | NULL             |
| 11 | 1      | 111-0001 | Data Science Basics | Non-fiction | 22.50 | A2            | 1           | Shinas Library         | 1            | 5.000000      | 2025-12-10       |
| 12 | 2      | 111-0002 | SQL Mastery         | Reference   | 35.00 | A3            | 1           | Shinas Library         | 1            | 4.000000      | 2025-12-12       |
| 13 | 3      | 111-0003 | Fairy Tales         | Children    | 12.00 | A4            | 1           | Shinas Library         | 1            | 5.000000      | 2025-12-15       |
| 14 | 4      | 111-0004 | Mystery Island      | Fiction     | 18.00 | A5            | 1           | Shinas Library         | 0            | 0.000000      | NULL             |
| 15 | 5      | 111-0005 | AI Future           | Non-fiction | 28.00 | A6            | 1           | Shinas Library         | 0            | 0.000000      | NULL             |
| 16 | 6      | 222-0001 | Football Legends    | Fiction     | 16.50 | B3            | 1           | Suhar Library          | 0            | 0.000000      | NULL             |
| 17 | 7      | 222-0002 | World History       | Non-fiction | 24.00 | B4            | 1           | Suhar Library          | 0            | 0.000000      | NULL             |
| 18 | 8      | 222-0003 | Encyclopedia Vol 1  | Reference   | 50.00 | B5            | 1           | Suhar Library          | 0            | 0.000000      | NULL             |
| 19 | 9      | 222-0004 | Kids Stories        | Children    | 10.00 | B6            | 1           | Suhar Library          | 0            | 0.000000      | NULL             |
| 20 | 10     | 222-0005 | Science Explained   | Reference   | 45.00 | B7            | 1           | Suhar Library          | 0            | 0.000000      | NULL             |



## Section 4: Stored Procedures

### 16. sp\_IssueBook

The screenshot displays the SQL Server Enterprise Manager interface. The top pane shows the SQL script for creating and executing the stored procedure `sp_IssueBook`. The script includes error handling, an insert statement into the `Loan_table`, an update statement to set `IsAvailable = 0` for the specified book, and a success message. The bottom pane shows the execution results, indicating that the book was issued successfully. The left pane shows the database structure, including the `LibraryDBproject` and its various components, with `dbo.sp_IssueBook` highlighted under the `Stored Procedures` folder.

```
BEGIN
    PRINT 'Error: Member has overdue loans.';
    RETURN;
END

/* 4. Issue Book */
INSERT INTO Loan_table (MemberID, BookID, LoanDate, DueDate, Status)
VALUES (@MemberID, @BookID, CAST(GETDATE() AS DATE), @DueDate, 'Issued');

/* 5. Update Book availability */
UPDATE Book
SET IsAvailable = 0
WHERE BookID = @BookID;

PRINT 'Success: Book issued successfully.';
END;
GO

EXEC sp_IssueBook
    @MemberID = 2,
    @BookID = 5,
    @DueDate = '2026-01-15';
```

110 %

Messages

Success: Book issued successfully.

Completion time: 2025-12-30T15:19:59.9842296+04:00

LibraryDBproject

- Database Diagrams
- Tables
- Views
- External Resources
- Synonyms
- Programmability
  - Stored Procedures
    - System Stored Procedures
    - dbo.sp\_IssueBook
    - dbo.sp\_ReturnBook
  - Functions

## 17. sp\_ReturnBook

```
SQLQuerynew.sql -...BTESAM\ebtes (59))* X
----17--
----Before returning the book---

-- Check the Loan table
SELECT LoanID, MemberID, BookID, LoanDate, DueDate, ReturnDate, Status
FROM Loan_table
WHERE LoanID = 3;

-- Check the Book table
SELECT BookID, Title, IsAvailable
FROM Book
WHERE BookID = (SELECT BookID FROM Loan_table WHERE LoanID = 3);

-- Check Payment table for existing payments
SELECT * FROM Payment
WHERE LoanID = 3;

----Expected output message-----

---- Execute stored procedure----
EXEC sp_ReturnBook
    @LoanID = 3,
    @ReturnDate = '2025-12-30';
```

110 %

Results Messages

|   | PaymentID | LoanID | PaymentDate | Amount | Method  |
|---|-----------|--------|-------------|--------|---------|
| 1 | 3         | 3      | 2025-02-25  | 6.00   | Cash    |
| 2 | 6         | 3      | 2025-12-30  | 20.00  | Pending |

## 18. sp\_GetMemberReport

```
SQLQuery2.sql - EB...BTESAM\ebtes (59)) * X
-- 1. Member Basic Info
SELECT MemberID, FullName, Email, PhoneNumber, MembershipStartDate
FROM MEMBER_TABLE
WHERE MemberID = @MemberID;

-- 2. Current loans (status Issued or Overdue)
SELECT L.LoanID, B.Title, L.LoanDate, L.DueDate, L.Status
FROM Loan_table L
JOIN Book B ON L.BookID = B.BookID
WHERE L.MemberID = @MemberID AND L.Status IN ('Issued', 'Overdue');

-- 3. Loan History with Return status
SELECT L.LoanID, B.Title, L.LoanDate, L.DueDate, L.ReturnDate, L.Status
FROM Loan_table L
JOIN Book B ON L.BookID = B.BookID
WHERE L.MemberID = @MemberID;

-- 4. Total fines paid and pending
SELECT
    SUM(CASE WHEN P.Method != 'Pending' THEN P.Amount ELSE 0 END) AS TotalFinesPaid,
    SUM(CASE WHEN P.Method = 'Pending' THEN P.Amount ELSE 0 END) AS TotalFinesPending
FROM Loan_table L
LEFT JOIN Payment P ON L.LoanID = P.LoanID
WHERE L.MemberID = @MemberID;
```

100 % 5 Review by member

Results Messages

|   | LoanID | Title               | LoanDate   | DueDate    | Status |  |
|---|--------|---------------------|------------|------------|--------|--|
| 1 | 1      | Data Science Basics | 2025-12-01 | 2025-12-15 | Issued |  |
| 2 | 11     | Fairy Tales         | 2025-12-30 | 2025-12-31 | Issued |  |

|   | LoanID | Title               | LoanDate   | DueDate    | ReturnDate | Status   |
|---|--------|---------------------|------------|------------|------------|----------|
| 1 | 1      | Data Science Basics | 2025-12-01 | 2025-12-15 | NULL       | Issued   |
| 2 | 2      | SQL Mastery         | 2025-12-02 | 2025-12-16 | 2025-12-14 | Returned |
| 3 | 11     | Fairy Tales         | 2025-12-30 | 2025-12-31 | NULL       | Issued   |

|   | TotalFinesPaid | TotalFinesPending |
|---|----------------|-------------------|
| 1 | 8.50           | 0.00              |

|   | ReviewID | Title               | Rating | Comments      | ReviewDate |
|---|----------|---------------------|--------|---------------|------------|
| 1 | 7        | Data Science Basics | 5      | Amazing read! | 2025-12-10 |
| 2 | 8        | SQL Mastery         | 4      | so happy.     | 2025-12-12 |

## 19. sp\_MonthlyLibraryReport

```
SQLQuery2.sql - EB...BTESAM\ebtes (59)) *
-- Total loans issued in that month
SELECT COUNT(*) AS TotalLoansIssued
FROM Loan_table L
JOIN Book B ON L.BookID = B.BookID
WHERE B.LibraryID = @LibraryID
AND L.LoanDate BETWEEN @StartDate AND @EndDate;

-- Total books returned in that month
SELECT COUNT(*) AS TotalBooksReturned
FROM Loan_table L
JOIN Book B ON L.BookID = B.BookID
WHERE B.LibraryID = @LibraryID
AND L.ReturnDate BETWEEN @StartDate AND @EndDate;

-- Total revenue collected in that month
SELECT ISNULL(SUM(P.Amount), 0) AS TotalRevenue
FROM Payment P
JOIN Loan_table L ON P.LoanID = L.LoanID
JOIN Book B ON L.BookID = B.BookID
WHERE B.LibraryID = @LibraryID
AND P.PaymentDate BETWEEN @StartDate AND @EndDate
AND P.Method != 'Pending';

-- Most borrowed genre in that month
SELECT TOP 1 B.Genre, COUNT(*) AS BorrowCount
FROM Loan_table L
JOIN Book B ON L.BookID = B.BookID
WHERE B.LibraryID = @LibraryID
AND L.LoanDate BETWEEN @StartDate AND @EndDate
GROUP BY B.Genre
ORDER BY BorrowCount DESC;

-- Top 3 most active members by number of loans in that month
SELECT TOP 3 M.MemberID, M.FullName, COUNT(*) AS LoanCount
FROM Loan_table L
JOIN Book B ON L.BookID = B.BookID
```

100 %

Results Messages

| TotalLoansIssued |   |
|------------------|---|
| 1                | 4 |

| TotalBooksReturned |   |
|--------------------|---|
| 1                  | 1 |

| TotalRevenue |      |
|--------------|------|
| 1            | 8.50 |

| Genre      | BorrowCount |
|------------|-------------|
| 1 Children | 2           |

| MemberID | FullName | LoanCount |
|----------|----------|-----------|
| 1 1      | EBTESAM  | 3         |
| 2 2      | AHMED    | 1         |

✓ Query executed successfully.

EBTESAM (

## 2. Documentation Document

Explanation of how your stored procedures handle edge cases:

| Stored Procedure        | Edge Case / Scenario               | How It's Handled                                                                          |
|-------------------------|------------------------------------|-------------------------------------------------------------------------------------------|
| sp_IssueBook            | Book does not exist                | Checks BookID in Book table; raises error 'Book not found.'                               |
|                         | Book is not available              | Checks IsAvailable = 1; if 0, returns error 'Book is not available.'                      |
|                         | Member has overdue loans           | Checks Loan_table for Status = 'Overdue'; prevents issuing another loan if overdue exists |
|                         | Successful loan                    | Inserts new Loan_table record, updates Book.IsAvailable = 0, returns success message      |
| sp_ReturnBook           | Loan record does not exist         | Checks LoanID in Loan_table; raises error 'Loan record not found.'                        |
|                         | Return date earlier than loan date | chk_ReturnDate constraint prevents invalid return dates                                   |
|                         | Book returned late                 | Calculates DaysOverdue; computes fine (DaysOverdue * 2)                                   |
|                         | Update book availability           | Sets Book.IsAvailable = 1                                                                 |
|                         | Successful return                  | Prints total fine amount (even if 0)                                                      |
| sp_GetMemberReport      | Member does not exist              | Returns empty result sets; no errors thrown                                               |
|                         | Member has no current loans        | Current loans result set is empty (LEFT JOIN ensures member still appears)                |
|                         | Member has no past loans           | Loan history result set is empty                                                          |
|                         | Member has no fines                | Total fines shows 0 using ISNULL                                                          |
|                         | Member has not written reviews     | Reviews result set is empty                                                               |
| sp_MonthlyLibraryReport | Library does not exist             | Returns empty or NULL values                                                              |
|                         | No loans in specified month        | Total loans and returned books show 0                                                     |
|                         | No payments/fines                  | Total revenue shows 0 using ISNULL                                                        |
|                         | No loans for a genre               | Most borrowed genre may return NULL                                                       |

## Important Notes

### 3Library:

SQLQuery\_COODING...ESAM\ebtes (74))\*

```
-- LIBRARY-Table---  
  
CREATE TABLE LIBRARY_Table (  
    LibraryID INT IDENTITY(1,1) PRIMARY KEY,  
    LName NVARCHAR(100) NOT NULL UNIQUE,  
    Lib_Location NVARCHAR(100) NOT NULL,  
    ContactNumber NVARCHAR(20) NOT NULL,  
    EstablishedYear INT  
);  
  
INSERT INTO LIBRARY_Table (LName, Lib_Location, ContactNumber, EstablishedYear)  
VALUES  
    ('Shinas Library', 'shinas_UTAS', '123-456-7890', 1990),  
    ('Suhar Library', 'UNIVERSITY', '987-654-3210', 2010),  
    ('Muscat Central Library', 'Muscat', '555-1111', 2005);  
  
SELECT * from LIBRARY_Table;
```

100 %

Results Messages

|   | LibraryID | LName                  | Lib_Location | ContactNumber | EstablishedYear |
|---|-----------|------------------------|--------------|---------------|-----------------|
| 1 | 1         | Shinas Library         | shinas_UTAS  | 123-456-7890  | 1990            |
| 2 | 2         | Suhar Library          | UNIVERSITY   | 987-654-3210  | 2010            |
| 3 | 3         | Muscat Central Library | Muscat       | 555-1111      | 2005            |



## 20 book:

| SQLQuery_COODING...ESAM\ebtes (74))* X                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |                |                     |             |       |             |               |           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------|---------------------|-------------|-------|-------------|---------------|-----------|
| REFERENCES LIBRARY_PUBLIS (LibraryID,<br>ON DELETE CASCADE<br>ON UPDATE CASCADE<br>);                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |                |                     |             |       |             |               |           |
| INSERT INTO Book (ISBN, Title, Genre, Price, ShelfLocation, LibraryID)<br>VALUES<br>-- Library 1<br>( '111-0001', 'Data Science Basics', 'Non-fiction', 22.50, 'A2', 1),<br>( '111-0002', 'SQL Mastery', 'Reference', 35.00, 'A3', 1),<br>( '111-0003', 'Fairy Tales', 'Children', 12.00, 'A4', 1),<br>( '111-0004', 'Mystery Island', 'Fiction', 18.00, 'A5', 1),<br>( '111-0005', 'AI Future', 'Non-fiction', 28.00, 'A6', 1),<br>-- Library 2<br>( '222-0001', 'Football Legends', 'Fiction', 16.50, 'B3', 2),<br>( '222-0002', 'World History', 'Non-fiction', 24.00, 'B4', 2),<br>( '222-0003', 'Encyclopedia Vol 1', 'Reference', 50.00, 'B5', 2),<br>( '222-0004', 'Kids Stories', 'Children', 10.00, 'B6', 2), |        |                |                     |             |       |             |               |           |
| 100 %                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |                |                     |             |       |             |               |           |
| Results Messages                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |                |                     |             |       |             |               |           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | BookID | ISBN           | Title               | Genre       | Price | IsAvailable | ShelfLocation | LibraryID |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1      | 978-0140449136 | HARRY POTTER        | Fiction     | 15.99 | 1           | A1            | 1         |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2      | 987-0131103627 | TOM AND JARRY       | Reference   | 40.50 | 1           | B2            | 1         |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3      | 369-0064400558 | FOOTBALL            | Fiction     | 12.75 | 1           | C3            | 2         |
| 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 4      | 258-0195153446 | SALLY History       | Non-fiction | 18.00 | 1           | D4            | 2         |
| 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 5      | 111-0001       | Data Science Basics | Non-fiction | 22.50 | 1           | A2            | 1         |
| 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 6      | 111-0002       | SQL Mastery         | Reference   | 35.00 | 1           | A3            | 1         |
| 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 7      | 111-0003       | Fairy Tales         | Children    | 12.00 | 1           | A4            | 1         |
| 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 8      | 111-0004       | Mystery Island      | Fiction     | 18.00 | 1           | A5            | 1         |
| 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 9      | 111-0005       | AI Future           | Non-fiction | 28.00 | 1           | A6            | 1         |
| 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 10     | 222-0001       | Football Legends    | Fiction     | 16.50 | 1           | B3            | 2         |
| 11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 11     | 222-0002       | World History       | Non-fiction | 24.00 | 1           | B4            | 2         |
| 12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 12     | 222-0003       | Encyclopedia Vol 1  | Reference   | 50.00 | 1           | B5            | 2         |
| 13                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 13     | 222-0004       | Kids Stories        | Children    | 10.00 | 1           | B6            | 2         |
| 14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 14     | 222-0005       | Science Explained   | Reference   | 45.00 | 1           | B7            | 2         |
| 15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 15     | 333-0001       | Modern Fiction      | Fiction     | 20.00 | 1           | C1            | 3         |
| 16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 16     | 333-0002       | Learning Python     | Reference   | 42.00 | 1           | C2            | 3         |
| 17                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 17     | 333-0003       | Healthy Living      | Non-fiction | 19.00 | 1           | C3            | 3         |
| 18                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 18     | 333-0004       | Children Fun Book   | Children    | 11.50 | 1           | C4            | 3         |
| 19                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 19     | 333-0005       | Advanced SQL        | Reference   | 48.00 | 1           | C5            | 3         |
| 20                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 20     | 333-0006       | Short Stories       | Fiction     | 14.00 | 1           | C6            | 3         |
| 21                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 21     | 333-0007       | History of Oman     | Non-fiction | 26.00 | 1           | C7            | 3         |
| 22                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 22     | 333-0008       | Math for Kids       | Children    | 9.50  | 1           | C8            | 3         |
| 23                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 23     | 333-0009       | Tech Trends         | Non-fiction | 30.00 | 1           | C9            | 3         |
| 24                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 24     | 333-0010       | Fantasy World       | Fiction     | 17.00 | 1           | C10           | 3         |

## 10member:

SQLQuery\_COODING...ESAM\ebtes (74)\* X

```
);
```

```
INSERT INTO MEMBER_TABLE (FullName, Email, PhoneNumber, MembershipStartDate)
VALUES
('EBTESAM', 'EBTESAM@GOOGLE.com', '333-1234', '2025-02-03'),
('AHMED', 'AHMED@OUTLOOK.com', '111-5678', '2022-09-04'),
('FATMA', 'FATMA@HOTMAIL.com', '222-5678', '2020-11-11'),
('ALI', 'ALI@gmail.com', '900-1111', '2021-01-01'),
('SARA', 'SARA@gmail.com', '900-2222', '2021-03-10'),
('MOHAMMED', 'MOHAMMED@gmail.com', '900-3333', '2022-05-20'),
('HIND', 'HIND@gmail.com', '900-4444', '2022-08-15'),
('KHALID', 'KHALID@gmail.com', '900-5555', '2023-01-01'),
('AISHA', 'AISHA@gmail.com', '900-6666', '2023-02-14'),
('YOUSEF', 'YOUSEF@gmail.com', '900-7777', '2023-04-01');

SELECT * from MEMBER_TABLE;
```

100 %

Results Messages

|    | MemberID | FullName | Email              | PhoneNumber | MembershipStartDate |
|----|----------|----------|--------------------|-------------|---------------------|
| 1  | 1        | EBTESAM  | EBTESAM@GOOGLE.com | 333-1234    | 2025-02-03          |
| 2  | 2        | AHMED    | AHMED@OUTLOOK.com  | 111-5678    | 2022-09-04          |
| 3  | 3        | FATMA    | FATMA@HOTMAIL.com  | 222-5678    | 2020-11-11          |
| 4  | 5        | ALI      | ALI@gmail.com      | 900-1111    | 2021-01-01          |
| 5  | 6        | SARA     | SARA@gmail.com     | 900-2222    | 2021-03-10          |
| 6  | 7        | MOHAMMED | MOHAMMED@gmail.com | 900-3333    | 2022-05-20          |
| 7  | 8        | HIND     | HIND@gmail.com     | 900-4444    | 2022-08-15          |
| 8  | 9        | KHALID   | KHALID@gmail.com   | 900-5555    | 2023-01-01          |
| 9  | 10       | AISHA    | AISHA@gmail.com    | 900-6666    | 2023-02-14          |
| 10 | 11       | YOUSEF   | YOUSEF@gmail.com   | 900-7777    | 2023-04-01          |

## 10 loan

SQLQuery2.sql - EB...BTESAM\ebtes (59))\* X

VALUES

```
(1, 1, '2025-12-01', '2025-12-15', NULL, 'Issued'),  
(1, 2, '2025-12-02', '2025-12-16', '2025-12-14', 'Returned'),  
(2, 3, '2025-12-05', '2025-12-20', NULL, 'Issued'),  
(4, 1, '2025-04-01', '2025-04-15', NULL, 'Issued'),  
(5, 2, '2025-05-03', '2025-04-18', NULL, 'Issued'),  
(6, 3, '2025-02-01', '2025-08-13', NULL, 'Issued'),  
(7, 2, '2025-07-08', '2025-07-12', NULL, 'Issued'),  
(8, 1, '2025-03-09', '2025-09-11', NULL, 'Issued'),  
(9, 1, '2025-06-03', '2025-10-10', NULL, 'Issued'),  
(10, 3, '2025-01-05', '2025-02-20', '2025-04-18', 'Returned');
```

SELECT \* from Loan\_table;

100 %

Results Messages

|    | LoanID | MemberID | BookID | LoanDate   | DueDate    | ReturnDate | Status   |
|----|--------|----------|--------|------------|------------|------------|----------|
| 1  | 1      | 1        | 1      | 2025-12-01 | 2025-12-15 | NULL       | Issued   |
| 2  | 2      | 1        | 2      | 2025-12-02 | 2025-12-16 | 2025-12-14 | Returned |
| 3  | 3      | 2        | 3      | 2025-12-05 | 2025-12-20 | NULL       | Issued   |
| 4  | 4      | 4        | 1      | 2025-04-01 | 2025-04-15 | NULL       | Issued   |
| 5  | 5      | 5        | 2      | 2025-05-03 | 2025-04-18 | NULL       | Issued   |
| 6  | 6      | 6        | 3      | 2025-02-01 | 2025-08-13 | NULL       | Issued   |
| 7  | 7      | 7        | 2      | 2025-07-08 | 2025-07-12 | NULL       | Issued   |
| 8  | 8      | 8        | 1      | 2025-03-09 | 2025-09-11 | NULL       | Issued   |
| 9  | 9      | 9        | 1      | 2025-06-03 | 2025-10-10 | NULL       | Issued   |
| 10 | 10     | 10       | 3      | 2025-01-05 | 2025-02-20 | 2025-04-18 | Returned |

## 4payment

```
SQLQuery2.sql - EB...BTESAM\ebtes (59))*  X
REFERENCES Loan_table(LoanID)
ON DELETE CASCADE
ON UPDATE CASCADE
);
INSERT INTO Payment (LoanID, PaymentDate, Amount, Method)
VALUES
(2, '2025-12-14', 5.00, 'Cash'),
(2, '2025-12-14', 3.50, 'Card'),
(3, '2025-02-25', 6.00, 'Cash'),
(8, '2025-05-20', 8.50, 'Card');

SELECT * from Payment;
```

100 %

Results Messages

|   | PaymentID | LoanID | PaymentDate | Amount | Method |
|---|-----------|--------|-------------|--------|--------|
| 1 | 1         | 2      | 2025-12-14  | 5.00   | Cash   |
| 2 | 2         | 2      | 2025-12-14  | 3.50   | Card   |
| 3 | 3         | 3      | 2025-02-25  | 6.00   | Cash   |
| 4 | 4         | 8      | 2025-05-20  | 8.50   | Card   |